# WITLE ACADEMY ERD

KGOMOTSO SHABANE

WitleShop (Pty) Ltd is a South African online retailer of electronics, clothing and home appliances. This document describes the database that manages customers, products, orders, payments, deliveries and suppliers.

## 1. Entities

The design has nine entities.

| # | Entity | Purpose | Type |
|---|---|---|---|
| 1 | CUSTOMER | A registered buyer | Strong |
| 2 | ADDRESS | A delivery address belonging to a customer | Dependent on CUSTOMER |
| 3 | CATEGORY | A group of products (e.g. Electronics) | Strong |
| 4 | SUPPLIER | A company that supplies products | Strong |
| 5 | PRODUCT | An item for sale | Strong |
| 6 | ORDERS | A purchase by one customer (named ORDERS because ORDER is a reserved SQL word) | Strong |
| 7 | ORDER_ITEM | The products contained in an order | Associative (bridge) |
| 8 | PAYMENT | The payment for an order | Strong |
| 9 | DELIVERY | The delivery of an order | Strong |

A customer can have several addresses, so they cannot be stored as one attribute of CUSTOMER.

An order can contain many products and a product can appear in many orders (many-to-many). This is split into two one-to-many relationships through ORDER_ITEM.

## 2. Attributes, Primary Keys and Foreign Keys

PK = primary key, FK = foreign key, UK = unique (no duplicates allowed).

| Entity | Attributes | Primary key | Foreign keys |
|---|---|---|---|
| CUSTOMER | customer_id, full_name, email (UK), phone_number, registration_date | customer_id | None |
| ADDRESS | address_id, customer_id, street_address, suburb, city, province, postal_code, is_default | address_id | customer_id |
| CATEGORY | category_id, category_name (UK), description | category_id | None |
| SUPPLIER | supplier_id, supplier_name, contact_email, contact_phone | supplier_id | None |
| PRODUCT | product_id, product_name, description, price, stock_quantity, category_id, supplier_id | product_id | category_id, supplier_id |
| ORDERS | order_id, customer_id, order_date, order_status, total_amount | order_id | customer_id |
| ORDER_ITEM | order_id, product_id, quantity, unit_price | order_id + product_id (composite) | order_id, product_id |
| PAYMENT | payment_id, order_id (UK), payment_date, payment_method, payment_status, amount_paid | payment_id | order_id |
| DELIVERY | delivery_id, order_id (UK), address_id, delivery_date, delivery_status, courier_name, tracking_number | delivery_id | order_id, address_id |

**Keys:** every table uses a system-generated integer ID as its primary key. ORDER_ITEM instead uses the combination of order_id and product_id, which also stops the same product being listed twice on one order.

## 3. Relationships and Cardinality

| Relationship | Type | Meaning | Foreign key |
|---|---|---|---|

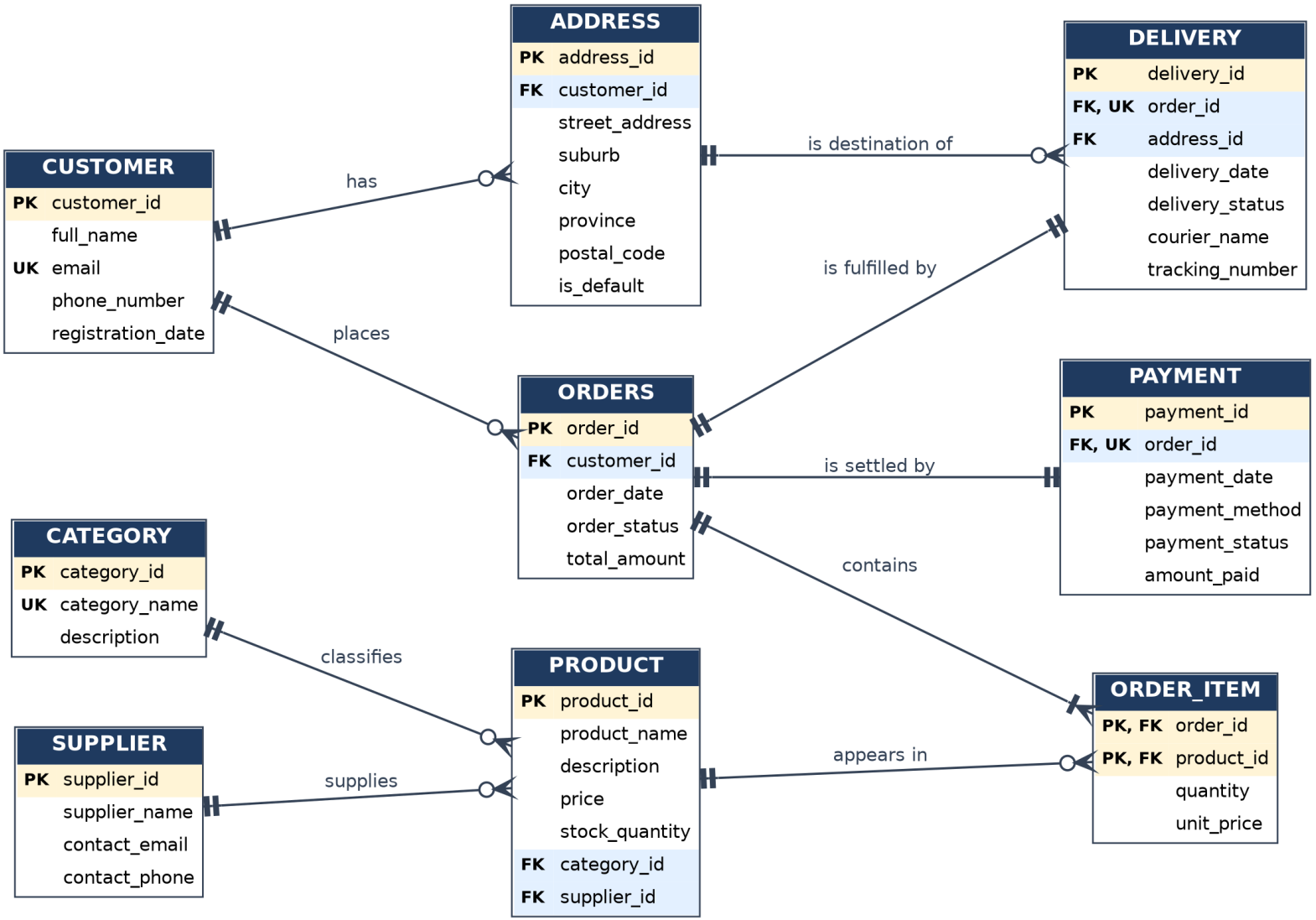
| | | | |
|---|---|---|---|
| CUSTOMER has ADDRESS | 1:M | One customer has many addresses; each address belongs to one customer | customer_id in ADDRESS |
| CUSTOMER places ORDERS | 1:M | One customer places many orders; each order belongs to one customer | customer_id in ORDERS |
| CATEGORY classifies PRODUCT | 1:M | One category has many products; each product has one category | category_id in PRODUCT |
| SUPPLIER supplies PRODUCT | 1:M | One supplier supplies many products; each product has one supplier | supplier_id in PRODUCT |
| ORDERS contains PRODUCT | M:N | An order has many products; a product is in many orders | Bridge table ORDER_ITEM |
| ORDERS is settled by PAYMENT | 1:1 | One order has one payment; one payment belongs to one order | order_id (UNIQUE) in PAYMENT |
| ORDERS is fulfilled by DELIVERY | 1:1 | One order has one delivery; one delivery belongs to one order | order_id (UNIQUE) in DELIVERY |
| ADDRESS is destination of DELIVERY | 1:M | One address can receive many deliveries; each delivery goes to one address | address_id in DELIVERY |

**Summary:** five one-to-many, two one-to-one and one many-to-many relationship. The many-to-many is implemented as two one-to-many relationships through ORDER_ITEM.

## 4. Business Rules in the Design

- **Registration first:** every order must belong to a CUSTOMER, and email must be unique.
- **One payment and one delivery per order:** enforced by a UNIQUE foreign key (order_id) in PAYMENT and DELIVERY.
- **Delivery address:** the address used for a delivery must belong to the customer who placed the order. This is checked when the delivery is saved (by the application or a database trigger).
- **Allowed values:** order_status is Pending, Shipped, Delivered or Cancelled; payment_method is Card, EFT or PayFast.
- **Prices and totals:** unit_price is stored in ORDER_ITEM so later price changes do not alter past orders; total_amount is stored because the brief lists it, but it can be calculated from the order items.
- **Assumed values:** the brief gives no payment or delivery statuses, so these are assumed.

## 5. Entity Relationship Diagram



## How to read the diagram

| Symbol | Meaning |
|---|---|
| Two short bars \|\| | Exactly one (mandatory) |
| Circle and crow's foot O< | Zero or many (optional, many) |
| Bar and crow's foot \|< | One or many (at least one) |
| PK / FK / UK | Primary key / foreign key / unique key |
| Verb on a line | Name of the relationship, e.g. "places" |